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RV College of Engineering, Bengaluru - 560059.

Title:AI-Driven Smart Civic Complaint Management System

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SDG:Sustainable Cities & Communities (SDG 11)

Introduction

Modern cities struggle to manage civic complaints efficiently due to fragmented reporting systems, manual triage, and lack of real-time visibility. Traditional grievance mechanisms fail to capture visual and spatial context, leading to delayed responses, inefficient resource allocation, and declining citizen trust.

Our Solution: An AI-powered, citizen-centric complaint reporting system that uses image and text intelligence to automatically classify, prioritize, and route civic issues in real time.

Problem Definition

The Challenge:

- Manual complaint triaging causes delays
- Image and location context is missing
- Departments are overloaded with unstructured complaints

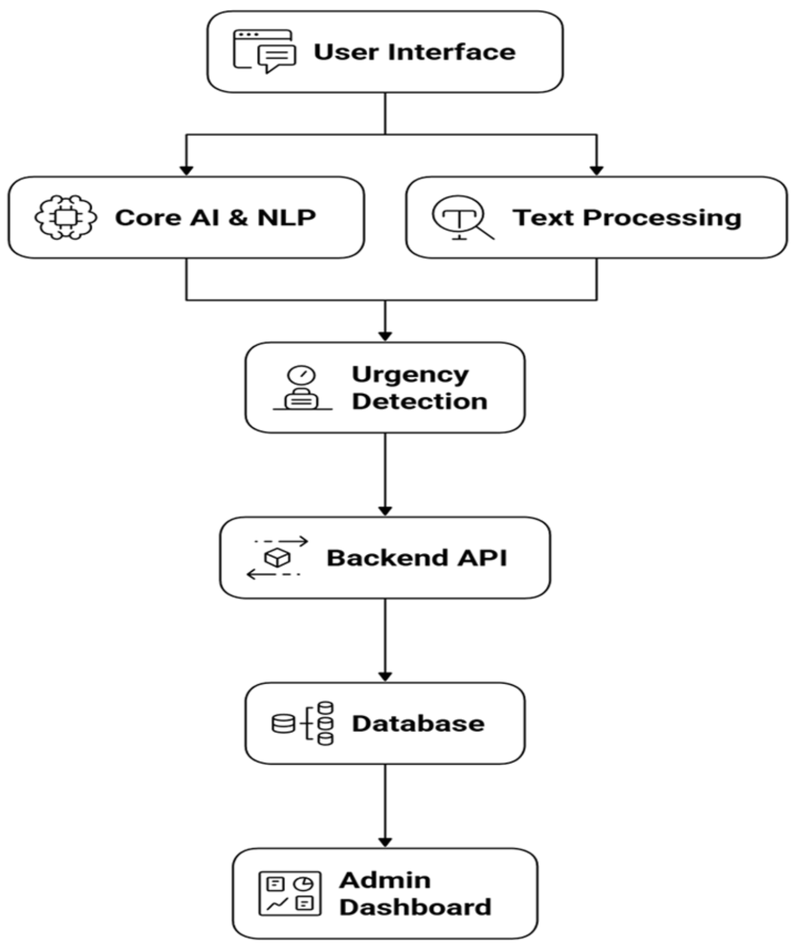
Why Traditional Methods Fail:

- Text-only complaints are ambiguous
- No automated routing
- Poor visibility for citizens and authorities

Objectives

- To design and implement a multi-modal complaint intake framework
- To develop an AI-based classification engine
- To establish a backend service architecture
- To construct an administrative workflow and analytics layer

Methodology



- **Multi-Modal Complaint Intake Framework**

We integrate Telegram interfaces through webhook APIs and implement a unified input-processing pipeline to accept text, media, and location data.

- **AI-Based Classification Engine**

We construct an NLP-driven classifier, supported optionally by a lightweight vision model, to interpret complaint content.

- **Backend Architecture and Data Management**

We develop RESTful backend services that manage complaint storage, authentication, status updates, and geotag data using a structured database schema. The backend forms the central communication layer between user interfaces, AI modules, and administrative tools.

- **Administrative Workflow and Analytics Layer**

We build a dashboard interface that visualizes complaints, clusters geo-data, and supports department-wise routing.

Tools used

AI Models:

- MobileNetV2 – Image-based complaint classification
- TF-IDF + Logistic Regression – Text complaint classification

Technologies:

- Telegram Bot for citizen interaction
- Flask REST API backend
- PostgreSQL + PostGIS for spatial data
- Heatmap-based admin dashboard

Results and Discussions

- High-Accuracy Vision Model: The MobileNetV2 classification engine achieved a weighted accuracy of 93.36% across 8 urban issue categories, with "Road Signs" achieving the highest F1-Score of 0.9882.
- Precision Urgency Detection: The text analysis module (TF-IDF & spaCy) delivered an F1-Score of 99.83%, successfully prioritizing keywords like "emergency" and "danger" even within informal complaint descriptions.
- Operational Efficiency: Utilizing a 0.75 confidence threshold, the system successfully auto-routed 85% of 3,056 simulated complaints to the correct departments without human intervention.
- Accelerated Resolution: The system drastically reduced processing times compared to the traditional 24–48 hour manual workflow utilized by current municipal methods.
- Societal Impact (SDG 11): By leveraging a familiar platform like Telegram, the system lowers reporting barriers, directly addressing the 78% of unresolved civic complaints in cities like Bengaluru.

Conclusions

SnapFix demonstrates how AI-driven participatory sensing can modernize civic infrastructure management.By merging familiar communication platforms with efficient AI models, the system offers a scalable, transparent, and inclusive alternative to conventional e-governance solutions.

Future Scope:

- Predictive maintenance using complaint trends
- Multilingual and voice-based complaint support
- Edge-AI deployment for faster inference
- Explainable AI for administrative decision support

Outcome of the work

The project resulted in the successful development and pilot deployment of SnapFix, an AI-powered Telegram bot for reporting urban infrastructure issues, achieving 93.36% image classification accuracy and 85% automated complaint routing. These outcomes highlight the project's readiness for academic publication in smart cities journals, such as npj Urban Sustainability, and demonstrate its potential for scalable real-world impact aligned with SDG 11.

References

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